

The Probability of a Run

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Abstract

We deduce the explicit formula for the probability of a run of r successes in n trials.

1 Introduction

A famous problem in classical probability was first stated in DE MOIVRE's treatise, *The Doctrine of Chances* [2] as *Problem LXXIV*:

“To find the Probability of throwing a Chance assigned a given number of times without intermission, in any given number of Trials.”

We formulate this more explicitly as follows:

In a series of independent trials, an event E has the constant probability p . If, in this series, E occurs at least r times in succession, we say that there is a **run** of r successes. *What is the probability of having a run of r successes in n trials, where naturally $n > r$?*

Let us denote by y_n the unknown probability of a run of r in n trials. The classical solution to the problem, (FELLER [3], USPENSKY [11]), consists of deducing the following *difference equation* for the *complementary* probability $z_n := 1 - y_n$:

$$\boxed{z_{n+1} - z_n + qp^r z_{n-r} = 0} \quad (1)$$

where $q := 1 - p$, and then concluding that the *generating function*:

$$G(x) := z_0 + z_1 x + z_2 x^2 + \cdots + z_n x^n + [\cdots] \quad (2)$$

is, in fact, the *rational* function

$$\boxed{G(x) = \frac{1 - p^r x^r}{1 - x + qp^r x^{r+1}}} \quad (3)$$

The coefficient of x^n gives the general formula for z_n . We will prove:

Theorem 1. *Let*

$$\beta_{n,r} := \sum_{l=0}^{\lfloor \frac{n}{r+1} \rfloor} (-1)^l \binom{n-lr}{l} (qp^r)^l \quad (4)$$

where $[\alpha] :=$ the greatest integer contained in α . Then

$$\boxed{z_n = \beta_{n,r} - p^r \beta_{n-r,r}} \quad (5)$$

This explicit formula for z_n , which one would expect to be at least as famous as the problem, itself, is amazingly hard to find in the literature. Although TODHUNTER [9] details the solutions of DE MOIVRE, CONDORCET, and LAPLACE, none of them gives the formula, although they all give versions of the generating function.

Nor is it to be found in the classical text of MARKOFF [6].

When we looked at the modern texts of CHUNG [1], FELLER [3], GNEDENKO [4], HOEL [5], PARZEN [7], ROSS [8], TUCKER [10], and USPENSKY [11], we were able to find only a *statement*, and that **without proof**, of **Theorem 1**, only in USPENSKY [11]. The most detailed presentation of the theory of runs on the internet is to be found in WEISSTEIN [12], but the explicit formula is not even mentioned there!

USPENSKY, ([11], page 79), states that the formula can be found “. . . according to the known rules.” Following FELLER ([3], pp. 275-276), if we write

$$G(x) = \frac{U(x)}{V(x)} \quad (6)$$

then the formula for z_n is:

$$z_n = \frac{\rho_1}{x_1^{n+1}} + \frac{\rho_2}{x_2^{n+1}} + \cdots + \frac{\rho_{r+1}}{x_{r+1}^{n+1}} \quad (7)$$

where $x_1, x_2, \dots, x_{r+1}$ are the $r+1$ distinct roots of $V(x) = 0$ and

$$\rho_k = \frac{-U(x_k)}{V'(x_k)}. \quad (8)$$

Unfortunately, the equation $V(x) = 0$ cannot, in general, be solved explicitly for its $r+1$ roots, and so “. . . the known rules” are useless in this case.

We therefore offer the following simple derivation of the formula for z_n , based on the binomial theorem.

2 Proof of Theorem 1

It suffices to prove the formula for $\beta_{n,r}$ since the formula for z_n is an immediate consequence of it. Now,

$$\begin{aligned}
\sum_{n=0}^{\infty} \beta_{n,r} x^n &\equiv \frac{1}{1 - x + qp^r x^{r+1}} \\
&= \frac{1}{1 - x(1 - qp^r x^r)} \\
&= \sum_{k=0}^{\infty} \{x(1 - qp^r x^r)\}^k \\
&= \sum_{k=0}^{\infty} x^k \sum_{l=0}^k \binom{k}{l} (-qp^r x^r)^l \\
&= \sum_{k=0}^{\infty} \sum_{l=0}^k (-1)^l \binom{k}{l} (qp^r)^l x^{r l + k}.
\end{aligned}$$

We must now determine the coefficient, $\beta_{n,r}$, of x^n in this series. Thus, for fixed r and n , we must find all pairs of integers (l, k) , with $0 \leq l \leq k$ which satisfy the equation:

$$rl + k = n, \tag{9}$$

since each such pair contributes the summand

$$(-1)^l \binom{k}{l} (qp^r)^l = (-1)^l \binom{n - lr}{l} (qp^r)^l \tag{10}$$

to the final value of $\beta_{n,r}$. By inspection we note that

$$\begin{aligned}
n &= r \cdot 0 + n \\
&= r \cdot 1 + (n - r) \\
&= r \cdot 2 + (n - 2r) \\
&= r \cdot 3 + (n - 3r) \\
&= \dots \dots \dots \\
&= (n - r) + r \cdot 1.
\end{aligned}$$

and these equations correspond to the pairs

$$(l, k) = (0, n), (1, n - r), (2, n - 2r), \dots, (n - r, r),$$

respectively. We do not include $(l, k) = (n, 0)$ since the corresponding summand has the value 0.

By (9) and the final equation in the list above, the largest value of l occurs when $l = k$ and thus satisfies the equation:

$$lr + r = n$$

and we conclude that

$$l = \left\lceil \frac{n}{r+1} \right\rceil. \quad (11)$$

Therefore, l takes on the values $0, 1, 2, \dots, \left\lceil \frac{n}{r+1} \right\rceil$, and the proof is complete.

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